VOLTWYN VW-MTC1 MazeTrack Controller — Rev A

Objective wire-maze assessment controller for ataxia research: 4-ch Hall sensing, 12-bit Hall ADC, IMU, 16-bit synchronous touch detection, haptics, Li-ion power, USB-C, microSD, Wi-Fi/BLE.



Power

File: Power.kicad_sch

Power: USB-C, charger, power path, 3.3 V buck-boost

MCU

File: MCU.kicad_sch

MCU: ESP32-S3, UI, IMU, storage, expansion

Sensing

File: Sensing.kicad_sch

Sensing: Hall front end, 12-bit ADC, handle I/O, haptics

MazeIO

File: MazeIO.kicad_sch

Maze I/O: wire / start / finish touch detection, buzzer

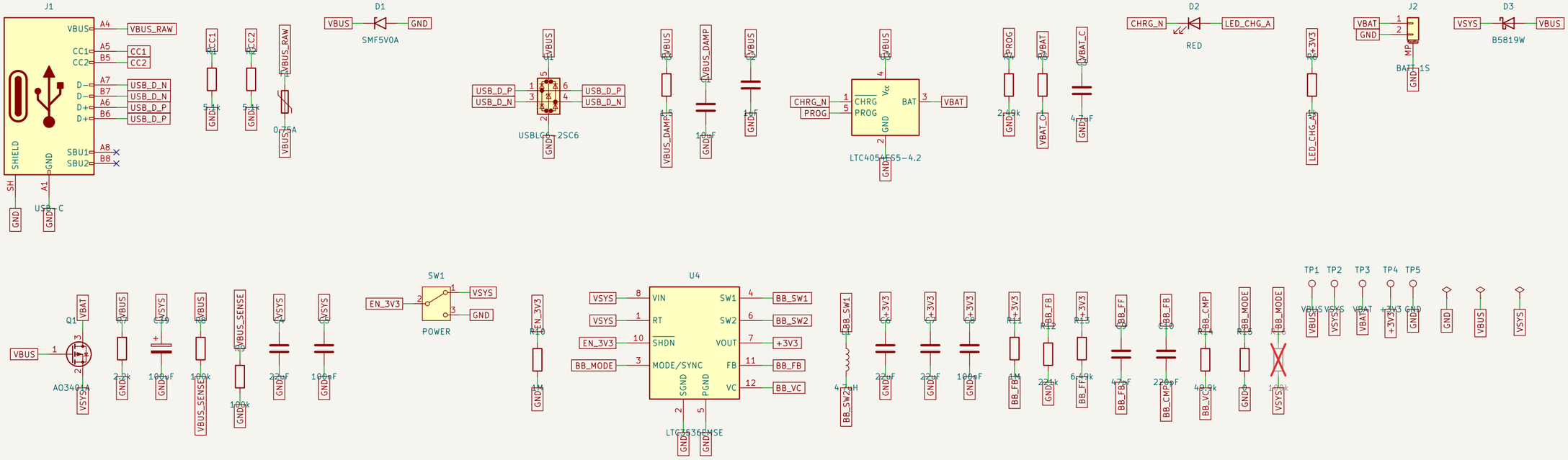
Designed by S. Veeriah. Generated from design.py netlist spec
VW-MTC1 MazeTrack Controller — Root
Voltwyn

Sheet: /
File: VW-MTC1.kicad_sch

Title: VW-MTC1 MazeTrack Controller — Top level

Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/5

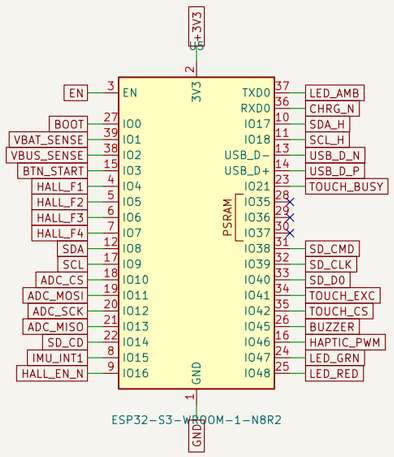
Power: USB-C, charger, power path, 3.3 V buck-boost

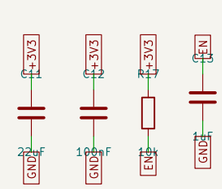


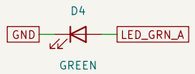
Designed by S. Veeriah. Generated from design.py netlist spec
VW-MTC1 MazeTrack Controller — Power
Voltwyn

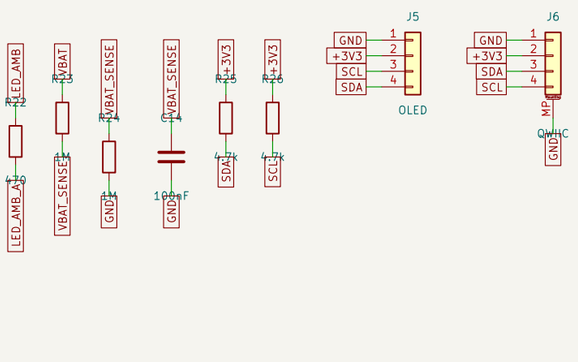
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File: Power.kicad_sch		
Title: Power: USB-C, charger, power path, 3.3 V buck-boost		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/5

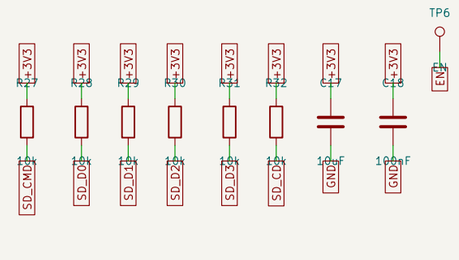
MCU: ESP32-S3, UI, IMU, storage, expansion











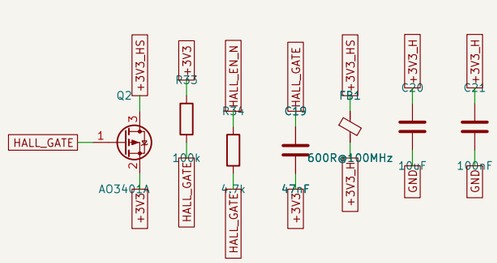
Designed by S. Veeriah. Generated from design.py netlist spec
VW-MTC1 MazeTrack Controller — MCU
Voltwyn

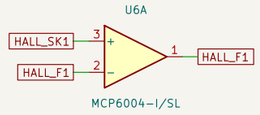
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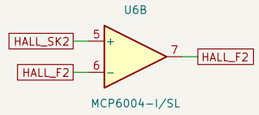
Title: MCU: ESP32-S3, UI, IMU, storage, expansion

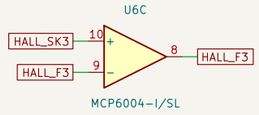
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KiCad E.D.A. 10.0.6		Id: 3/5

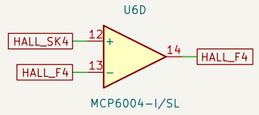
Sensing: Hall front end, 12-bit ADC, handle I/O, haptics

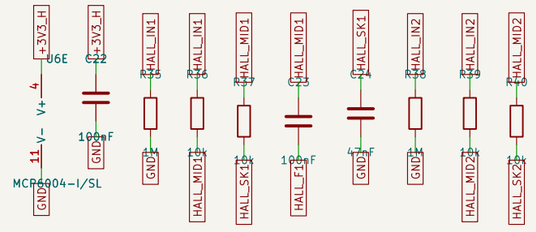


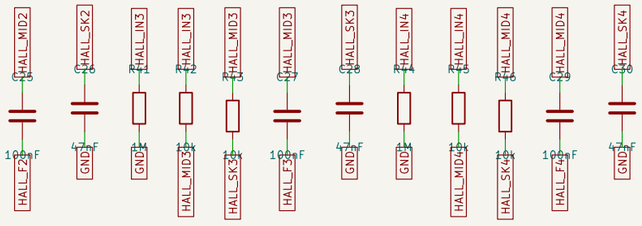


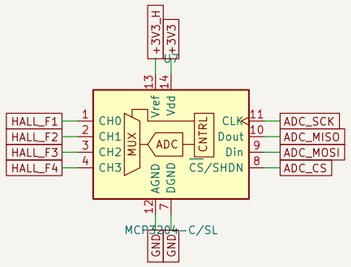


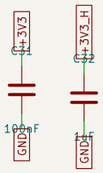


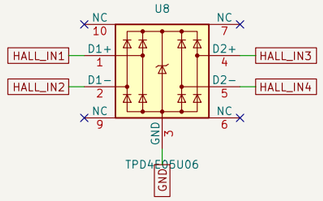


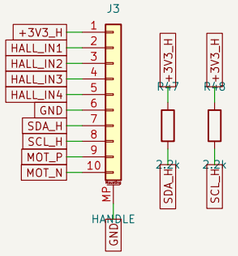


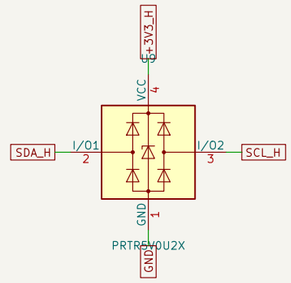


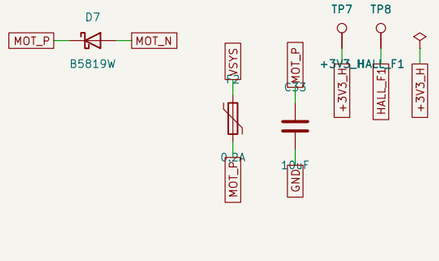












Designed by S. Veeriah. Generated from design.py netlist spec
VW-MTC1 MazeTrack Controller — Sensing
Voltwyn

Sheet: /Sensing/
File: Sensing.kicad_sch

Title: Sensing: Hall front end, 12-bit ADC, handle I/O, haptics

Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 4/5

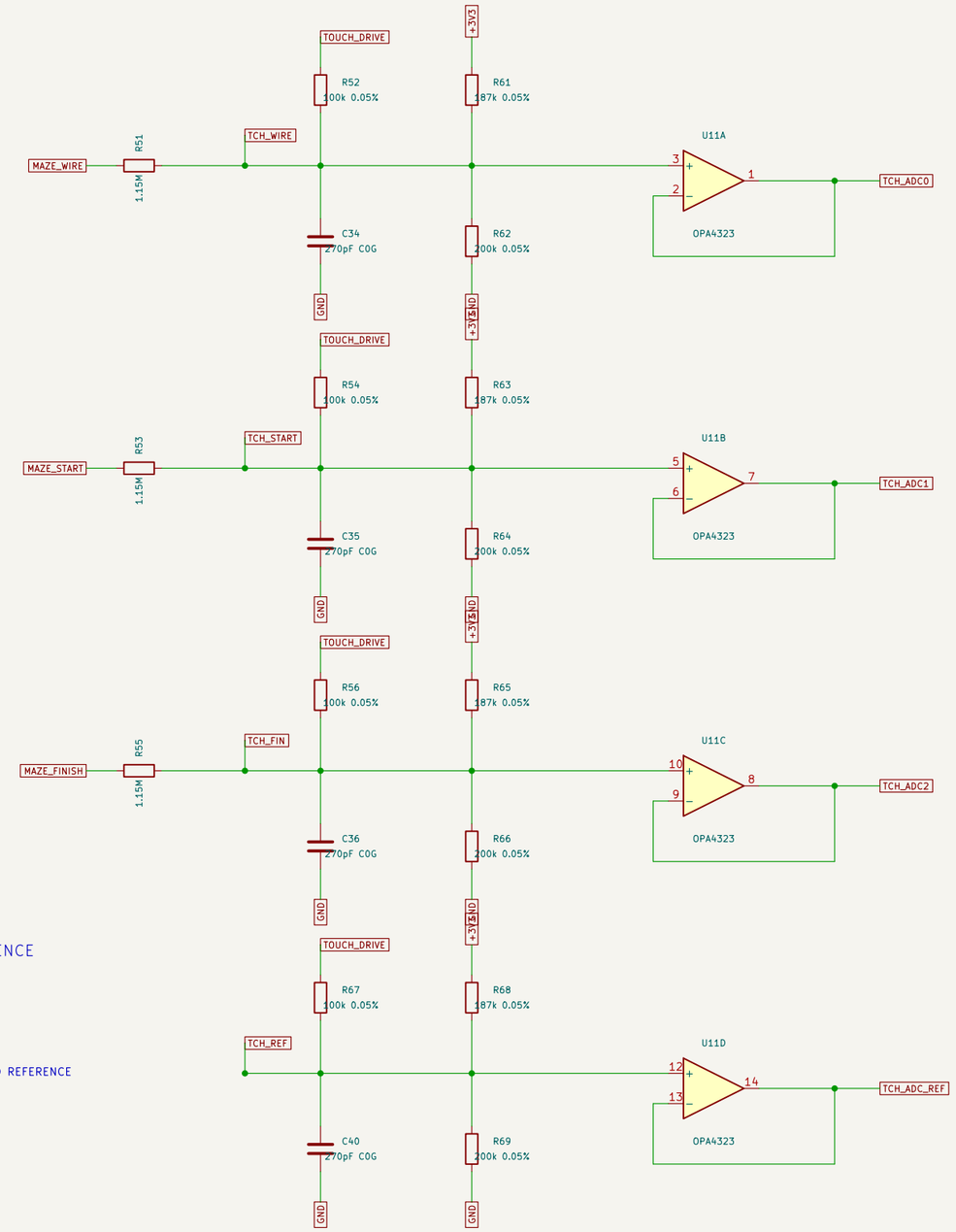
WIRE

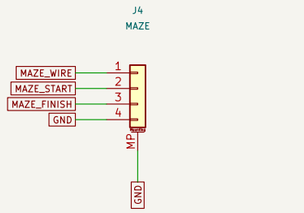
START

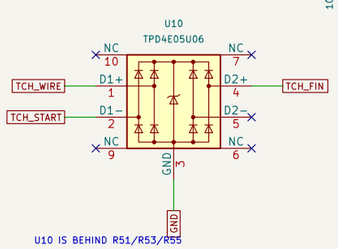
FINISH

REFERENCE

UNLOADED REFERENCE

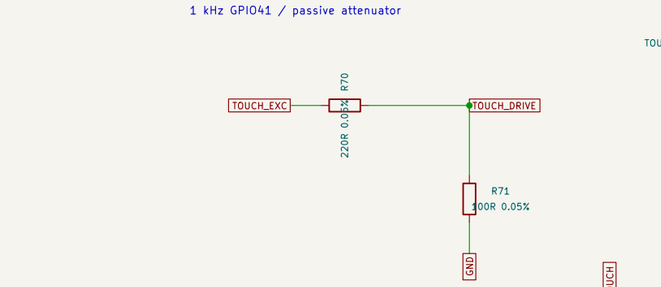


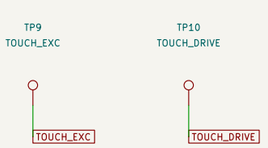


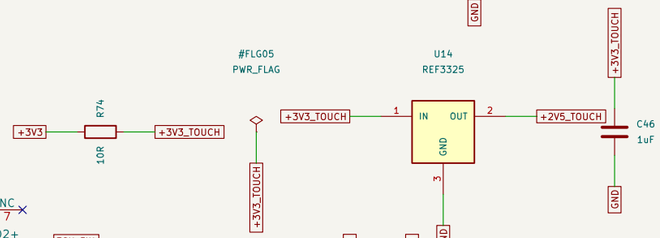


U10 IS BEHIND R51/R53/R55

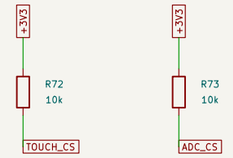
1 kHz GPIO41 / passive attenuator

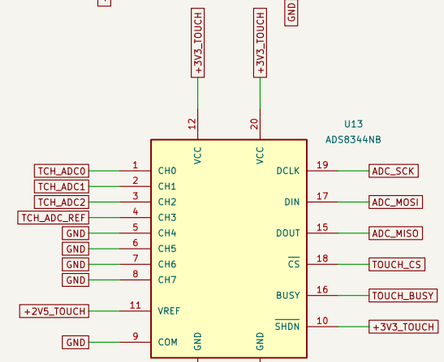


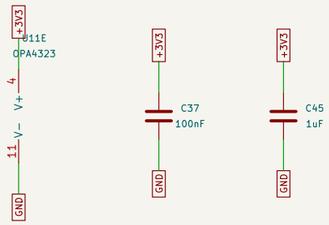




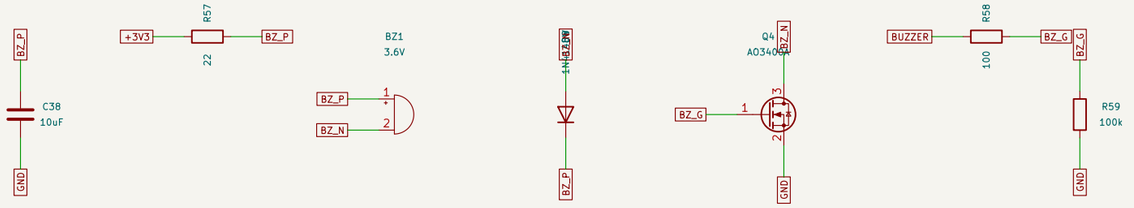
Both ADCs deselected at reset







BUZZER — circuit unchanged



Rev B: 5 ms gaps; 1.15 MOhm source limit per input.
Current acceptance and cable-capacitance envelope: see Rev01 design note.
Use N8R2 module. GPIO41 = EXC, GPIO42 = CS, GPIO21 = BUSY.

Rev B touch redesign; see verification record and measurement boundary
VW-MTC1 MazeTrack Controller — MazeIO
Voltwyn
Sheet: /MazeIO/
File: MazeIO.kicad_sch

Title: Synchronous touch acquisition and buzzer

Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 5/5